

Is the predictability in risk or in direction? A pre-registered evaluation of regime-conditioned analogue forecasting in financial markets

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Abstract

Analogue (nearest-neighbour) forecasting is popular in trading because each forecast is backed by concrete historical episodes, but it is easy to evaluate wrongly. We build a leakage-safe analogue engine—with an exclusion zone that keeps analogue outcomes independent, a Markov mixture over volatility regimes and labels produced by the same trade simulator that executes the strategy—and ask where analogue information is predictive. After a development study on nine assets, we pre-registered eight hypotheses and tested them on fifteen new cryptocurrencies, equity indices, commodities and currencies. All eight replicated. Analogues carry no information about the direction of prices: information coefficients are within ± 0.06 and no strategy survives a deflated-Sharpe test. They are informative about risk: combined with HAR they are statistically indistinguishable from HAR and stay in the model confidence set for 23 of 25 confirmatory series, while gradient-boosted trees given the same inputs are markedly worse; kernel weighting improves the analogue forecaster by 5%; analogue-conditioned Value-at-Risk is as well calibrated as filtered historical simulation; and volatility-managed exposure reduced the maximum drawdown for all 24 assets without raising Sharpe ra-

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tios. A static Poisson crash gate of the kind used in trading software is worse than a constant forecast in 31 of 40 series, and a Hawkes process improves on it but not on a logistic model of current volatility. Code, data scripts and the pre-registration are public.

Keywords: Analogue forecasting, Nearest neighbours, Volatility forecasting, Value-at-Risk, Hawkes process, Pre-registration, Model confidence set

1. Introduction

Forecasting by analogy is one of the oldest ideas in prediction. Lorenz (1969) asked how well the atmosphere could be forecast by finding past states that resemble today’s and following what happened next; Farmer and Sidorowich (1987) turned the idea into nearest-neighbour prediction of chaotic time series. In finance, analogue (nearest-neighbour) methods have been applied to exchange rates (Fernández-Rodríguez et al., 1999) and are now common in retail trading software, where their appeal is interpretability: each forecast rests on a list of concrete historical episodes that a user can inspect, in contrast to the opaque internal states of deep networks.

This paper began as an attempt to publish such a system: a trading engine that matched the current 20-bar return against history with a one-dimensional k -nearest-neighbour rule, adjusted the resulting win rate with a two-state Markov chain on volatility, suspended trading when a Poisson estimate of the probability of a large price jump exceeded a threshold, and sized positions with a fractional Kelly criterion (Kelly, 1956). Its in-sample backtests on Bitcoin and gold looked favourable. Rebuilding it for a rigorous evaluation changed the question. Analogue methods are easy to evaluate wrongly: an analogue whose outcome window has not closed leaks the future; adjacent bars have near-identical features, so the k nearest analogues are often one episode counted k times; and trying many configurations and reporting the best inflates performance (Bailey and López de Prado, 2014; Harvey et al., 2016). Once these problems are removed, the natural question is not whether one configuration is profitable but *where*, if anywhere, analogue information is predictive.

We answer that question with a leakage-safe analogue engine and a two-stage design. A *development* study on nine assets (three hourly cryptocurrencies, six daily equity, bond, gold and currency series) was used for all model building. Its findings were then turned into eight hypotheses and *pre-*

registered—the hypotheses, models, asset list and analysis code were committed to a public repository and tagged before any data for the confirmatory assets was downloaded—and tested on fifteen new assets. Pre-registration is standard in experimental sciences (Nosek et al., 2018) and has been advocated for empirical finance (Harvey, 2017), but it remains rare in forecasting studies.

The analogue engine is used for four targets: the direction of a trade (the original use), the realised variance of returns over the next h bars, the probability of a large jump, and the one-day return distribution (Value-at-Risk and Expected Shortfall). Each is compared with standard benchmarks under a walk-forward protocol: HAR (Corsi, 2009) and GARCH (Bollerslev, 1986) for volatility, Hawkes processes (Hawkes, 1971) for jumps, filtered historical simulation (Barone-Adesi et al., 1999) and GARCH- t for tail risk, and gradient-boosted trees (Friedman, 2001) as a flexible machine-learning comparator.

Our contributions are threefold. First, we provide a pre-registered, multi-asset evaluation of analogue forecasting across direction and risk, with the model confidence set (Hansen et al., 2011), Diebold–Mariano tests (Diebold and Mariano, 1995), the Fissler–Ziegel loss for VaR/ES (Fissler and Ziegel, 2016; Patton et al., 2019) and deflated Sharpe ratios (Bailey and López de Prado, 2014). Second, we introduce design elements that make analogue forecasts statistically meaningful: an exclusion zone that keeps analogue outcome windows independent, a mixture over volatility regimes weighted by the Markov forecast of the regime over the horizon, and labels produced by the same trade simulator that executes the strategy. Third, we release the engine and every script as open-source software, so that each table in this paper is reproduced by one command.

The results are summarised in Section 5. In brief, analogue methods carry no information about direction, but they produce risk forecasts that are competitive with the best standard models and, for the tails, unusually well calibrated.

2. Related literature

Analogue and nearest-neighbour forecasting. After Lorenz (1969) and Farmer and Sidorowich (1987), nearest-neighbour predictors were applied to financial series under the hypothesis that nonlinear deterministic structure could be exploited where linear models fail. Fernández-Rodríguez et al. (1999)

found that simultaneous nearest-neighbour predictors improved on random-walk forecasts of European exchange rates. The weak-form efficient-markets hypothesis (Fama, 1970) predicts that such directional structure should be absent or small relative to trading costs. Kernel regression (Nadaraya, 1964; Watson, 1964) is the smooth counterpart of k NN averaging and appears here as a variant of the engine.

Volatility forecasting. Volatility clustering makes risk far more predictable than returns. GARCH(1,1) (Bollerslev, 1986) is hard to beat in large comparisons (Hansen and Lunde, 2005); the heterogeneous autoregressive (HAR) model (Corsi, 2009) captures long memory with three averages of past variance. Comparing volatility forecasts requires loss functions that are robust to noise in the realised proxy; QLIKE and MSE on the variance scale are robust in this sense (Patton, 2011). The model confidence set (Hansen et al., 2011) identifies the set of models that cannot be distinguished from the best.

Jumps and self-excitation. Large price moves cluster. Hawkes processes (Hawkes, 1971) model this through an intensity that jumps after each event and decays; they are widely used for market microstructure (Bacry et al., 2015) and for contagion between markets (Aït-Sahalia et al., 2015). The original system used a static Poisson rate estimated over a long window, which cannot represent clustering.

Tail-risk forecasting. Filtered historical simulation (Barone-Adesi et al., 1999) rescales historical standardised residuals by a current volatility forecast. Coverage of VaR forecasts is assessed with the tests of Kupiec (1995) and Christoffersen (1998). Expected Shortfall is not elicitable on its own but is jointly elicitable with VaR (Fissler and Ziegel, 2016), and the FZ0 loss of Patton et al. (2019) is the standard tool for comparing (VaR, ES) forecasts.

Economic value and overfitting. Scaling exposure inversely to forecast variance improves risk-adjusted returns in some samples (Moreira and Muir, 2017), but out-of-sample gains are fragile (Cederburg et al., 2020). Back-tests of many configurations must be deflated for multiple testing (Bailey and López de Prado, 2012, 2014; Harvey et al., 2016).

3. Methodology

3.1. The analogue engine

Let $x_t \in \mathbb{R}^m$ be a causal embedding of the state at bar t and $y_j \in \mathbb{R}^p$ a vector of targets attached to anchor j , fully observed at $j + \delta$. At bar t the candidate set is $\mathcal{C}_t = \{j : j \leq t - \delta\}$, so no candidate's outcome extends beyond t . Distances are Euclidean, $d_{tj} = \|x_t - x_j\|$.

Exclusion zone. Candidates are accepted in increasing distance, but a candidate is skipped if it lies within e bars of an already accepted analogue. With $e \geq h$ the outcome windows of accepted analogues do not overlap, so their outcomes are not mechanically dependent. Without this rule the nearest neighbours of a smooth feature are usually adjacent bars from one episode. A configuration guard rejects settings in which $2ke$ exceeds a quarter of the available history, because the greedy rule would then be forced to accept distant, non-analogous candidates.

Regime mixture. Each bar has a volatility state $s_t \in \{0, 1\}$ (ATR above or below its long moving average, following Wilder, 1978). Analogues are selected separately within each regime, giving group means $\bar{y}_0(t)$ and $\bar{y}_1(t)$. A two-state Markov chain estimated on past transitions gives the probability $\pi_1(t)$ of being in the high state, averaged over the next h steps; the forecast is $\hat{y}_t = (1 - \pi_1)\bar{y}_0 + \pi_1\bar{y}_1$ with standard error $[(1 - \pi_1)^2\hat{\sigma}_0^2/k_0 + \pi_1^2\hat{\sigma}_1^2/k_1]^{1/2}$. This replaces the ad hoc adjustment of the original system, which shrank the analogue win rate toward one half by the persistence probability, with an application of the law of total probability over regimes (cf. Hamilton, 1989).

Kernel variant. Instead of k neighbours, all candidates can be weighted by $w_{tj} = \exp(-d_{tj}^2/2b^2)$; standard errors then use the Kish (1965) effective sample size.

3.2. Direction model

The embedding is the last $w = 20$ log returns summed into m segments and scaled by the EWMA volatility at t ; $m = 1$ without scaling is the feature of the original system. The target is not the sign of a future return but the net R-multiple of the exact trade the strategy would take at the anchor: entry at the next open, stop $2.5 \times \text{ATR}$, target $1.5 \times$ the stop, exit after $h = 20$ bars, costs deducted. The same simulator executes live signals, so labels and execution cannot disagree. A long (short) trade is entered when the

lower confidence bound $\hat{R} - z$ se is positive and exceeds the opposite side's; the fraction of equity risked is the fractional Kelly quantity $\frac{1}{2} E[R]/E[R^2]$, capped at 2%.

3.3. Volatility model

The target is $\log RV_{t,h}$ with $RV_{t,h} = h^{-1} \sum_{u=1}^h r_{t+u}^2$ and $h = 24$ for hourly and $h \in \{5, 22\}$ for daily series. The analogue embedding is the *shape* of recent volatility: the log mean squared return in $m = 4$ consecutive segments of h bars, each relative to a long-run EWMA variance $\bar{\sigma}_t^2$. The analogues forecast the log ratio of future variance to $\bar{\sigma}_t^2$. We also report *Analogue+HAR*, the equal-weight average of the analogue and HAR log forecasts. Log-scale forecasts are converted to the variance scale with the mean exponentiated residual of the fitting window.

Benchmarks are EWMA (J.P. Morgan/Reuters, 1996) with half-life chosen by QLIKE on the fitting window from $\{5, 10, 20, 50, 100, 200\}$; GARCH(1,1) with variance targeting and its closed-form h -step mean; log-HAR with averages over (1, 5, 22) days or (1, 24, 168) hours; and gradient-boosted trees (histogram-based, 300 iterations, learning rate 0.05, no tuning) given the union of the HAR inputs, the analogue embedding, the regime and the long-run level.

3.4. Tail events

A raw jump occurs at t when $|r_t| > 3\bar{\sigma}_{t-1}$, measured against the long-run rather than the current volatility so that volatility clustering is not normalised away. The target is whether at least one jump occurs in the next h bars. Forecasts: the fitting-window base rate; the static Poisson estimate of the original system, $1 - \exp(-\hat{\lambda}h)$ with $\hat{\lambda}$ the jump rate over the last 1,000 bars; an exponential Hawkes process with intensity $\lambda(t) = \mu + \alpha\beta \sum_{t_i < t} e^{-\beta(t-t_i)}$, fitted by maximum likelihood, giving $1 - \exp\{-\mu h - \alpha S_t(1 - e^{-\beta h})\}$ with S_t the decayed event count; a logistic regression on $\log(\hat{\sigma}_{\text{EWMA},t}^2/\bar{\sigma}_t^2)$; and the analogue frequency.

3.5. Value-at-Risk and Expected Shortfall

For one-day VaR and ES at $\alpha \in \{1\%, 2.5\%, 5\%\}$ all models except historical simulation (HS, 250 days) use the same GARCH(1,1) forecast $\hat{\sigma}_{t+1}$, isolating the shape of the innovation distribution: Gaussian (GARCH-N), standardised Student- t with maximum-likelihood degrees of freedom (GARCH- t), the empirical distribution of all past standardised residuals (FHS), and

the weighted empirical distribution of the standardised residuals that followed the k nearest volatility-shape analogues (Analogue). *Blend* averages the Analogue and FHS quantiles and shortfalls.

3.6. Volatility-managed exposure

A long position is scaled by $\min(\sigma^*/\hat{\sigma}_t, 1)$, where σ^* is the average volatility in the calibration period, with no leverage, a rebalancing band of 0.1 and costs of 5 bp per unit of turnover. A variant additionally sets exposure to zero when the Hawkes jump probability exceeds its calibration-period 90th percentile, reproducing the “lock” of the original system.

4. Experimental design

Data. Hourly spot prices from Binance and daily prices from Yahoo Finance (Table 1). The development set comprises BTC, ETH and PAXG (a gold-backed token) hourly from 2018 (PAXG from 2020), and the S&P 500 (from 1927), Nasdaq-100 ETF, gold ETF, long-Treasury ETF, EUR/USD and NIFTY 50 daily. The confirmatory set, fixed in the pre-registration, comprises SOL, BNB, XRP, ADA and LTC hourly and DAX, Nikkei 225, FTSE 100, Hang Seng, Russell 2000 ETF, emerging-markets ETF, silver ETF, oil ETF, GBP/USD and USD/JPY daily. Daily Yahoo series pass a documented filter that removes isolated bad prints (a move of more than eight 20-day standard deviations reversed by more than 70% on the next bar).

Walk-forward protocol. Each series is split into a calibration period (the first 30% of bars, and at least the warm-up plus 500 bars) and an out-of-sample period of five consecutive folds. For each fold, every fitted quantity—EWMA half-life, GARCH, HAR, gradient boosting, Hawkes and logistic parameters, bias corrections, and the direction model’s configuration chosen from an 18-point grid—uses only data observed before the fold. All results are out-of-sample.

Statistics. Volatility forecasts are compared with QLIKE and log-MSE, Diebold–Mariano tests with Newey–West variance at lag h (Newey and West, 1987), and the 90% model confidence set with the $T_{\max}$ statistic and 1,000 stationary-bootstrap draws (Politis and Romano, 1994). Jump forecasts use log-loss, Brier score and AUC. VaR/ES forecasts use Kupiec and Christoffersen tests, tick loss and the FZ0 loss, with a model confidence

Table 1: Data. Out-of-sample (OOS) bars follow the calibration period (volatility study, $h = 24$ hourly, $h = 5$ daily).

Set	Series	Ticker	Source	Freq.	Start	End	Bars	OOS bars
Development	BTC	BTCUSDT	Binance	1h	2018-01-01	2026-09-30	76,544	53,581
	ETH	ETHUSDT	Binance	1h	2018-01-01	2026-09-30	76,544	53,581
	PAXG	PAXGUSDT	Binance	1h	2020-08-28	2026-09-30	53,354	37,348
	S&P 500	^GSPC	Yahoo	1d	1927-12-30	2026-09-29	24,802	17,362
	Nasdaq-100	QQQ	Yahoo	1d	1999-03-10	2026-09-28	6,931	4,852
	Gold	GLD	Yahoo	1d	2004-11-18	2026-09-28	5,498	3,849
	Treasuries	TLT	Yahoo	1d	2002-07-30	2026-09-28	6,080	4,256
	EUR/USD	EURUSD=X	Yahoo	1d	2003-12-01	2026-09-28	5,920	4,144
	NIFTY 50	^NSEI	Yahoo	1d	2007-09-17	2026-09-29	4,670	3,115
Confirmatory	SOL	SOLUSDT	Binance	1h	2020-08-11	2026-09-30	53,775	37,643
	BNB	BNBUSDT	Binance	1h	2018-01-01	2026-09-30	76,551	53,586
	XRP	XRPUSDT	Binance	1h	2018-05-04	2026-09-30	73,625	51,538
	ADA	ADAUSDT	Binance	1h	2018-04-17	2026-09-30	74,037	51,826
	LTC	LTCUSDT	Binance	1h	2018-01-01	2026-09-30	76,551	53,586
	DAX	^GDAXI	Yahoo	1d	1987-12-30	2026-09-29	9,799	6,860
	Nikkei 225	^N225	Yahoo	1d	1965-01-05	2026-09-28	15,174	10,622
	FTSE 100	^FTSE	Yahoo	1d	1984-01-03	2026-09-29	10,797	7,558
	Hang Seng	^HSI	Yahoo	1d	1986-12-31	2026-09-28	9,807	6,865
	Russell 2000	IWM	Yahoo	1d	2000-05-26	2026-09-29	6,624	4,637
	Emerging mkts	EEM	Yahoo	1d	2003-04-14	2026-09-29	5,903	4,133
	Silver	SLV	Yahoo	1d	2006-04-28	2026-09-29	5,137	3,582
	Oil	USO	Yahoo	1d	2006-04-10	2026-09-29	5,150	3,595
	GBP/USD	GBPUSD=X	Yahoo	1d	2003-12-01	2026-09-28	5,935	4,155
	USD/JPY	JPY=X	Yahoo	1d	1996-10-30	2026-09-28	7,756	5,430

set on FZ0. Trading results use Sharpe ratios, drawdowns, the deflated Sharpe ratio for the direction grid, and paired stationary-bootstrap tests of Sharpe differences.

Pre-registration. After the development study, eight hypotheses (Table 2), the confirmatory asset list, an exclusion rule (fewer than 20,000 hourly or 3,000 daily bars), the unchanged model code and the analysis script were committed to the public repository and tagged **prereg-v1** before any confirmatory data was downloaded. Tests are across series; family-wise error over the eight primary endpoints is controlled with Holm’s method (Holm, 1979). Development-set results are exploratory—some were inspected during model building—and are never pooled with confirmatory results.

5. Results

Table 2 and Figure 1 summarise the pre-registered tests. Every development-set finding that was registered as a hypothesis replicated on the fifteen confirmatory assets, with no deviation from the registered plan and no series excluded. The subsections below report each part of the study, development and confirmatory sets side by side.

Table 2: Pre-registered hypotheses. Development statistics are exploratory; confirmatory tests use the registered code on fifteen new assets. Holm p adjusts over H1–H8.

ID	Hypothesis	Development (exploratory)		Confirmatory (pre-registered)		
		Statistic	p	Statistic	p	Holm p
H1	Analogue direction forecasts have zero mean IC across assets (null expected)	mean IC -0.0043; max DSR 0.58	0.562	mean IC +0.0031; max DSR 0.46	0.603	1.000
H2	Analogue+HAR QLIKE differs from HAR	median ratio 1.012	0.303	median ratio 1.001	0.672	1.000
H3	Analogue alone QLIKE differs from HAR	median ratio 1.103	0.001	median ratio 1.087	0.000	0.000
H4	Gaussian-kernel analogue beats kNN analogue (one-sided)	median ratio 0.967	0.000	median ratio 0.953	0.000	0.000
H5	Gradient boosting QLIKE differs from HAR	median ratio 1.522	0.000	median ratio 1.423	0.000	0.000
H6	Hawkes tail forecasts beat static Poisson (one-sided log-loss)	median ratio 0.930	0.000	median ratio 0.847	0.000	0.000
H7	Analogue-FHS blend FZ0 differs from FHS at alpha = 2.5%	median diff -0.0003	0.910	median diff -0.0005	0.934	1.000
H8	Volatility targeting (Analogue+HAR) reduces max drawdown vs buy & hold (one-sided sign test)	9/9 improve	0.002	15/15 improve	0.000	0.000

5.1. Direction: no predictability

Table 3 reports the information coefficient (IC; Spearman correlation between the forecast and the realised R-multiple of the trade), the Brier skill of the direction probability, and out-of-sample Sharpe ratios. Across 24 assets the IC never exceeds 0.06 in absolute value, the mean IC is -0.004 in the development set and 0.003 in the confirmatory set, and H1 is confirmed ($p = 0.60$). Richer embeddings ($m = 4$) do not help. The bar-for-bar replication of the original strategy (v1) and the walk-forward-selected v2 both trail buy-and-hold on almost every asset with positive drift, and the best deflated Sharpe ratio over the 18-configuration grid is 0.58 (development) and 0.46 (confirmatory), far below the 0.95 needed to claim skill. Positive Brier skill appears only for Nasdaq-100 and gold (development) and oil (confirmatory); all three had large changes in drift between the calibration and out-of-sample periods, which a fixed climatological benchmark cannot track, and none has a significant IC.

The original system’s claim that aggregating to higher frequencies acts as a “low-pass filter” receives only weak support: for Bitcoin the IC rises from 0.004 (15-minute bars) to 0.006 (hourly) and 0.02–0.035 (4-hour), but the 4-hour sample contains roughly a thousand independent 20-bar outcomes, and an IC of 0.03 is within sampling error.

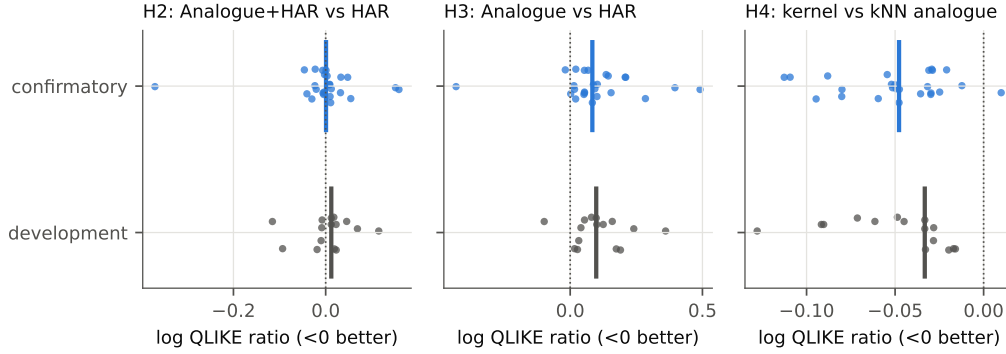


Figure 1: Per-series log QLIKE ratios for H2–H4 (points) and their medians (bars). Negative values favour the first-named model.

5.2. Volatility: competitive in combination

Table 4 summarises 40 series–horizon pairs (full results in Appendix Appendix A). The analogue forecaster alone is 11% worse than HAR on average and significantly worse (H3). Combined with HAR it is statistically indistinguishable from HAR (H2; median ratio 1.001 in the confirmatory set) and remains in the 90% model confidence set in 23 of 25 confirmatory series, compared with 24 for HAR and 20 for GARCH(1,1); these two counts are unchanged when bootstrap block lengths are chosen automatically (Appendix Appendix B). Gradient-boosted trees given a superset of the same information are the worst model, with median QLIKE 42–52% above HAR, and are in the confidence set for 2 of 40 series (H5): with a few thousand effectively independent observations the flexible model overfits the noisy volatility proxy, while the analogue estimator’s averaging over whole outcome windows limits its variance.

Figure 2 shows that the combination’s advantage over HAR is not driven by a few episodes: on Bitcoin the cumulative QLIKE gain rises steadily from 2024, while on the S&P 500 the combination is ahead of HAR for most of seven decades but loses ground in the 1987 and 2020 crashes. GARCH loses heavily to HAR on Bitcoin but leads on the S&P 500, in line with the view that no single volatility model dominates (Hansen and Lunde, 2005).

Ablations. Table 5 varies one design element at a time. The exclusion zone and the full history each improve QLIKE by 2–3%, confirming that independent outcomes and many candidates matter. The Markov regime mixture

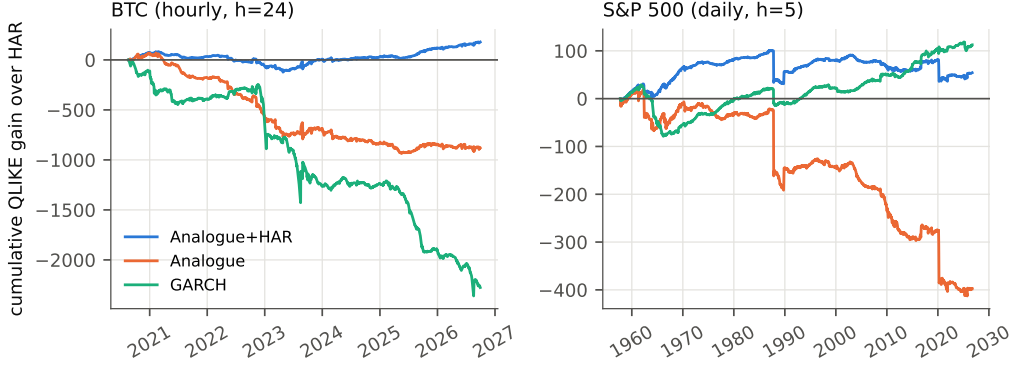


Figure 2: Cumulative QLIKE gain over HAR (development set). Rising lines beat HAR.

has no measurable effect. The Gaussian kernel is 5% better than k NN in both sets; this was found after the development protocol was fixed, registered as H4, and confirmed ($p < 10^{-6}$). Kernel weighting is therefore the recommended default for future work.

5.3. Tail events: clustering is volatility

Table 6 evaluates forecasts of at least one raw jump in the next h bars. The static Poisson rule of the original system is *worse than a constant forecast* in 11 of 15 development and 20 of 25 confirmatory series: a 1,000-bar average of past jump counts lags turning points in risk. A Hawkes process fixes this (H6; median log-loss ratio 0.85 versus Poisson), but a one-variable logistic regression on the EWMA variance is better still, with the lowest log-loss in most series. Jump clustering is therefore largely volatility clustering, and a Hawkes intensity adds little once the current variance is known. Figure 3 shows that the Poisson rule is also miscalibrated: its highest-risk bin has a lower realised jump frequency than its middle bins.

5.4. Value-at-Risk and Expected Shortfall: calibrated but not sharper

Table 7 compares one-day VaR/ES forecasts at 1%, 2.5% and 5%. Historical simulation and Gaussian GARCH fail coverage tests most often. The analogue-conditioned residual distribution passes the Kupiec test as often as FHS (69% of confirmatory cases for both) but ranks lower on the FZ0 loss: 250 analogues per regime give a noisier estimate of a 1% quantile than the full residual history. Averaging the two (Blend) recovers FHS-level accuracy

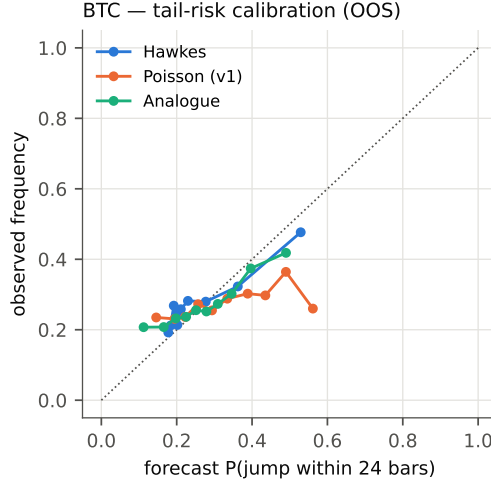


Figure 3: Reliability of jump-probability forecasts for Bitcoin (out-of-sample deciles).

but does not improve on it (H7), and FHS, Blend and GARCH- t form the top group in both sets.

5.5. Economic value: lower drawdowns, not higher Sharpe ratios

Scaling a long position by the Analogue+HAR volatility forecast reduced the maximum drawdown for all 9 development and all 15 confirmatory assets (H8; median reduction 5.6 percentage points in the confirmatory set) at the cost of slightly lower Sharpe ratios (median change -0.015 ; positive for 5 of 15 assets). This is the pattern documented by Cederburg et al. (2020): volatility management reliably reduces risk but does not reliably improve risk-adjusted returns out of sample. With dividend-adjusted prices for the four distribution-paying funds, Sharpe ratio levels rise but every comparison keeps its sign (Appendix Appendix B). The Hawkes “lock-out” of the original system lowered the Sharpe ratio further for 9 of 15 confirmatory assets, and significantly so for Bitcoin and Ethereum in the development set, because the periods after jumps include sharp rebounds (Table 8, Figure 4).

5.6. Interpretability and computation

Every analogue forecast can be decomposed into its analogues. Figure 5 shows the hour of the FTX collapse in November 2022: the ten nearest historical volatility shapes, what followed them, and the forecast. Here the

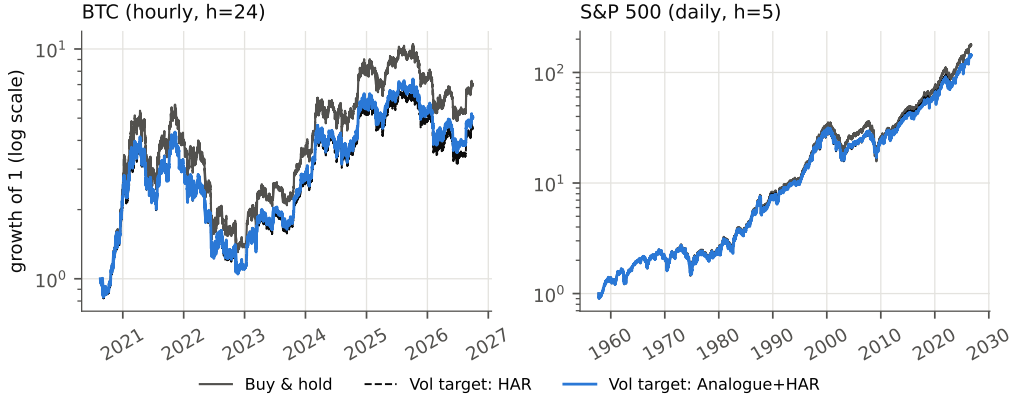


Figure 4: Growth of one unit, out-of-sample (development set).

analogues predicted a partial reversion of volatility that did not occur, which is the kind of error a user can see and reason about. The engine forecasts about 30,000 bars per second with a capped history and 200,000 bars in about 90 seconds with the full-history search on a 10-core laptop; the complete confirmatory study ran in under half an hour.

6. Discussion

Where the predictability is. The analogue engine extracts no information about the direction of prices but a substantial amount about their risk. This mirrors the broader distinction between the unpredictability of returns (Fama, 1970) and the strong predictability of volatility (Bollerslev, 1986; Corsi, 2009), and it holds across cryptocurrencies, equity indices, commodities, bonds and currencies, at hourly and daily frequency.

Implications for the original system. Of the original system's components, the directional analogue signal and the Poisson crash gate do not survive out-of-sample testing: the former has no skill, and the latter is worse than a constant. The regime mixture, which replaced an ad hoc adjustment, is harmless but adds nothing measurable. What survives is the core idea of matching the present to the past, redirected from direction to risk, where it is competitive with the standard models and yields well-calibrated tail forecasts with a transparent justification.

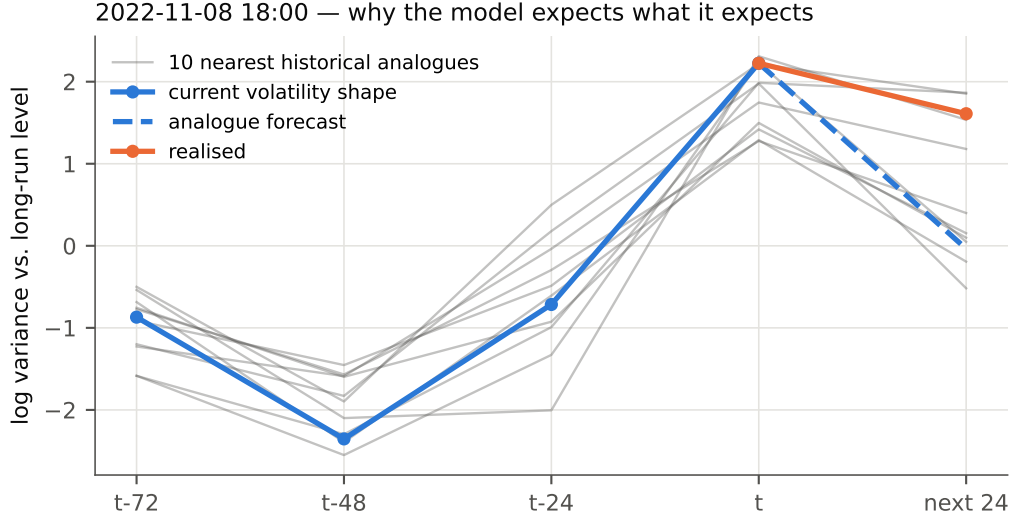


Figure 5: A forecast and its analogues: BTC, 8 November 2022, 18:00 UTC.

Why pre-registration mattered. Two development findings could have been overstated without the confirmatory test. The analogue VaR was the best-calibrated model in development but only tied with FHS on new assets; the kernel variant, found after the development design was fixed, might have been dismissed as a lucky ablation but replicated with $p < 10^{-6}$. Pre-registration separated these cases at negligible cost.

Limitations. Realised variance is computed from bar returns rather than intraday data, which adds noise to the target for daily series. The Analogue+HAR combination uses fixed equal weights. The embedding uses only a single series; cross-asset analogues are a natural extension. Transaction-cost assumptions are stylised, and the overlay ignores financing and cash returns. The main tables use unadjusted closing prices, as registered; Appendix Appendix B shows that dividend adjustment does not change the conclusions.

7. Conclusion

A pre-registered evaluation on 24 assets shows that analogue forecasting, a popular and interpretable tool, has no directional value in financial markets but produces risk forecasts that are competitive with HAR and GARCH,

calibrated tail forecasts, and reliable drawdown reductions. Its most useful design elements are an exclusion zone that keeps analogues independent, the full history, and kernel weighting. We release the engine and the pre-registration so that other analogue methods can be tested the same way.

Appendix A. Full volatility results

Bold entries are in the 90% model confidence set.

Series (horizon)	EWMA	GARCH	HAR	GBM	Analogue	Analogue+HAR
<i>Development set</i>						
BTC (h=24)	1.101	1.107	1.000	1.270	1.041	0.992
ETH (h=24)	1.147	1.067	1.000	1.251	1.033	0.990
EUR/USD (h=22)	0.987	1.078	1.000	1.399	1.017	0.911
EUR/USD (h=5)	0.939	0.928	1.000	1.525	1.028	0.982
Gold (h=22)	0.896	1.003	1.000	1.527	0.906	0.891
Gold (h=5)	1.015	0.969	1.000	1.625	1.103	1.012
NIFTY 50 (h=22)	1.044	0.894	1.000	2.016	1.272	1.071
NIFTY 50 (h=5)	1.073	1.004	1.000	1.574	1.106	1.023
Nasdaq-100 (h=22)	1.262	0.980	1.000	1.974	1.434	1.122
Nasdaq-100 (h=5)	1.022	0.942	1.000	2.139	1.084	1.018
PAXG (h=24)	1.109	1.087	1.000	1.308	1.172	1.047
S&P 500 (h=22)	1.222	1.023	1.000	1.466	1.209	1.023
S&P 500 (h=5)	1.075	0.984	1.000	1.152	1.056	0.992
Treasuries (h=22)	1.049	1.042	1.000	1.522	1.190	1.019
Treasuries (h=5)	1.010	0.975	1.000	1.368	1.133	1.012
Mean ratio	1.063	1.006	1.000	1.541	1.119	1.007
In 90% MCS	11/15	14/15	14/15	0/15	8/15	14/15
<i>Confirmatory set</i>						
ADA (h=24)	1.105	1.071	1.000	1.323	1.014	0.977
BNB (h=24)	1.151	1.162	1.000	1.242	1.070	1.001
DAX (h=22)	1.181	0.972	1.000	1.256	1.329	1.056
DAX (h=5)	1.044	0.962	1.000	1.281	1.055	0.993
Emerging mkts (h=22)	1.241	0.990	1.000	1.307	1.167	1.033
Emerging mkts (h=5)	1.051	0.984	1.000	1.358	1.098	1.012
FTSE 100 (h=22)	1.037	0.927	1.000	1.695	1.146	0.998
FTSE 100 (h=5)	0.993	0.936	1.000	1.410	1.017	0.980
GBP/USD (h=22)	0.983	0.991	1.000	1.624	1.108	1.009
GBP/USD (h=5)	1.000	0.982	1.000	1.645	1.087	1.011
Hang Seng (h=22)	1.017	0.950	1.000	1.756	1.231	1.034
Hang Seng (h=5)	1.011	0.986	1.000	1.423	1.091	1.008
LTC (h=24)	1.114	1.106	1.000	1.330	1.055	0.997
Nikkei 225 (h=22)	1.125	1.130	1.000	1.598	1.155	1.003
Nikkei 225 (h=5)	1.118	1.078	1.000	1.348	1.056	0.994
Oil (h=22)	1.362	0.986	1.000	2.214	1.486	1.164
Oil (h=5)	1.081	1.061	1.000	1.889	1.021	0.971
Russell 2000 (h=22)	1.333	0.924	1.000	1.808	1.634	1.172
Russell 2000 (h=5)	1.011	0.946	1.000	1.959	1.107	1.010
SOL (h=24)	1.105	1.110	1.000	1.313	1.053	0.996
Silver (h=22)	0.976	0.927	1.000	2.720	1.234	1.049
Silver (h=5)	0.979	0.970	1.000	1.600	1.002	0.960
USD/JPY (h=22)	0.658	1.058	1.000	1.432	0.650	0.691
USD/JPY (h=5)	1.012	0.977	1.000	1.375	1.021	0.978
XRP (h=24)	1.235	1.000	1.000	1.249	0.982	0.955
Mean ratio	1.077	1.007	1.000	1.566	1.115	1.002
In 90% MCS	18/25	20/25	24/25	2/25	12/25	23/25

Appendix B. Robustness

Dividend-adjusted prices. The study uses Yahoo’s unadjusted closing prices, as registered. Price indices (S&P 500, DAX and the like) have no adjusted series, and the gold, silver and oil funds pay no distributions, so adjustment matters only for the Nasdaq-100, long-Treasury, Russell 2000 and emerging-markets funds. Re-running the volatility and overlay experiments on their dividend-adjusted prices (Table B.9) moves QLIKE ratios by at most 0.026. Sharpe ratio levels rise for both buy-and-hold and volatility targeting—most for Treasuries, where coupons are the bulk of the return—but every comparison keeps its sign: volatility targeting still lowers the maximum drawdown and still slightly lowers the Sharpe ratio for three of the four funds.

Bootstrap block lengths. Model confidence sets and Sharpe-difference tests use the stationary bootstrap with fixed mean block lengths ($2h$; ten days for VaR/ES). Replacing them by the automatic choice of Politis and White (2004), with the correction of Patton et al. (2009), leaves the headline counts unchanged—Analogue+HAR remains in the 90% confidence set for 14 of 15 development and 23 of 25 confirmatory series, HAR for 14 and 24—and changes 19 of 672 confidence-set verdicts and 3 of 168 test verdicts (Table B.10). The automatic lengths are slightly more conservative, so the changes mostly admit additional models to a confidence set.

Kernel effective sample size. In kernel mode all anchors receive weight, and anchors whose outcome windows overlap are not independent, so the Kish (1965) effective sample size counted per anchor overstates the information in the analogue set. This affects the standard error of the kernel forecast and its minimum-sample rule, not its mean, so H4, which compares forecast accuracy, is unaffected. The released code offers an overlap-aware effective sample size that counts each block of h anchors as one observation; it is recommended for new work.

Data and code availability

All code, the pre-registration and every result table are available at <https://github.com/ManasDasri/analogue-risk> under the MIT license. The pre-registration is the tagged commit `prereg-v1`. Price data are downloaded from public Binance and Yahoo Finance endpoints by the included scripts.

Declaration of competing interest

The authors hold a patent on the original trading method evaluated in this paper. The evaluation was pre-registered and its results are reported in full.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to assist with software development, analysis and drafting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Table 3: Direction. IC: out-of-sample Spearman correlation between the analogue forecast and the realised trade R-multiple. Sharpe ratios at 5 bp per side. DSR: deflated Sharpe ratio of the walk-forward-selected v2 over its 18-configuration grid.

Asset	IC (long)		Brier skill	Sharpe ratio			DSR
	<i>m</i> =1	<i>m</i> =4		B&H	v1	v2 WF	
<i>Development set</i>							
BTC	0.006	0.028	-0.028	0.84	-0.44	-0.12	0.01
ETH	0.006	0.014	-0.029	0.78	-1.04	-0.76	0.00
EUR/USD	-0.014	-0.027	-0.025	-0.15	-0.35	-0.43	0.00
Gold	-0.040	0.012	0.077	0.71	0.34	0.40	0.42
NIFTY 50	-0.058	-0.030	-0.051	0.62	-0.00	-0.64	0.00
Nasdaq-100	-0.032	-0.027	0.025	0.96	0.25	0.44	0.58
PAXG	-0.007	0.005	-0.030	0.95	-0.85	-1.14	0.00
S&P 500	-0.006	0.005	-0.057	0.55	0.10	0.08	0.32
Treasuries	0.019	-0.018	-0.029	-0.16	0.03	-0.29	0.00
<i>Confirmatory set</i>							
ADA	0.011	0.018	-0.031	0.67	-0.51	-0.43	0.00
BNB	-0.001	0.014	-0.027	1.13	-0.65	-0.06	0.01
DAX	0.002	-0.007	-0.047	0.37	0.17	-0.12	0.01
Emerging mkts	0.006	-0.024	-0.053	0.30	-0.51	-0.12	0.04
FTSE 100	-0.001	0.021	-0.068	0.27	0.14	-0.05	0.01
GBP/USD	0.013	-0.023	-0.026	-0.12	0.21	-0.08	0.06
Hang Seng	0.005	0.001	-0.078	0.25	0.12	0.02	0.06
LTC	0.001	0.009	-0.032	0.46	-1.28	-0.44	0.00
Nikkei 225	0.009	0.035	-0.047	0.32	0.11	-0.02	0.03
Oil	-0.034	-0.052	0.113	0.34	0.03	-0.11	0.02
Russell 2000	-0.007	0.016	-0.052	0.52	-0.42	0.00	0.06
SOL	-0.010	-0.001	-0.037	0.82	-1.33	-0.92	0.00
Silver	0.006	-0.002	-0.048	0.49	-0.20	-0.74	0.00
USD/JPY	-0.007	0.017	-0.040	0.18	-0.02	-0.20	0.01
XRP	0.004	0.027	-0.028	0.81	-0.95	0.66	0.46

Table 4: Volatility forecasting summary (QLIKE relative to HAR; lower is better).

	EWMA	GARCH	HAR	GBM	Analogue	Analogue+HAR
<i>Development set (15 series)</i>						
Mean QLIKE / HAR	1.063	1.006	1.000	1.541	1.119	1.007
Median QLIKE / HAR	1.049	1.003	1.000	1.522	1.103	1.012
In 90% MCS	11	14	14	0	8	14
Lowest QLIKE	0	7	4	0	0	4
<i>Confirmatory set (25 series)</i>						
Mean QLIKE / HAR	1.077	1.007	1.000	1.566	1.115	1.002
Median QLIKE / HAR	1.051	0.986	1.000	1.423	1.087	1.001
In 90% MCS	18	20	24	2	12	23
Lowest QLIKE	1	14	2	0	1	7

Table 5: Ablations: mean QLIKE relative to the full k NN analogue model (< 1 : the variant is better).

	<i>m=1 (level only)</i>	<i>m=2</i>	<i>m=8</i>	<i>no regime mixture</i>	<i>no exclusion zone</i>	<i>Gaussian kernel (bw 0.6)</i>	<i>history capped at 5000</i>
Development	0.994	0.984	1.007	1.007	1.021	0.952	1.020
Confirmatory	0.990	0.980	1.022	0.990	1.030	0.952	1.023

Table 6: Jump forecasts: probability of at least one raw jump ($|r| > 3\bar{\sigma}$) in the next h bars.

	Constant	Poisson (v1)	Hawkes	Logistic-EWMA	Analogue
<i>Development set</i>					
Mean log-loss / constant	1.000	1.034	0.944	0.904	0.926
Mean AUC	0.432	0.490	0.570	0.670	0.656
Best log-loss (series)	0	0	0	12	3
Worse than constant	0	11	1	0	2
<i>Confirmatory set</i>					
Mean log-loss / constant	1.000	1.061	0.901	0.893	0.911
Mean AUC	0.437	0.502	0.613	0.710	0.679
Best log-loss (series)	0	1	6	16	2
Worse than constant	0	20	0	1	2

Table 7: One-day VaR/ES. Pass rates are the share of series $\times \alpha$ cases with $p > 0.05$.

	HS	GARCH-N	GARCH-t	FHS	Analogue	Blend
<i>Development set</i>						
Mean hit rate/ $\alpha - 1$	0.231	0.267	0.164	0.162	0.144	0.144
Kupiec pass rate	0.63	0.59	0.78	0.85	0.89	0.89
Christoffersen CC pass rate	0.41	0.56	0.67	0.74	0.81	0.74
Mean FZ0 rank (1 = best)	5.81	3.85	1.89	2.30	4.74	2.41
In 90% MCS (FZ0)	0.52	0.85	1.00	1.00	0.63	1.00
<i>Confirmatory set</i>						
Mean hit rate/ $\alpha - 1$	0.243	0.338	0.228	0.176	0.184	0.162
Kupiec pass rate	0.51	0.42	0.58	0.69	0.69	0.67
Christoffersen CC pass rate	0.31	0.44	0.60	0.71	0.67	0.71
Mean FZ0 rank (1 = best)	5.47	4.07	2.36	2.31	4.47	2.33
In 90% MCS (FZ0)	0.40	0.71	0.87	1.00	0.73	0.98

Table 8: Volatility-managed long exposure (no leverage, 5 bp per unit turnover). p : paired stationary-bootstrap test of the Sharpe difference with buy-and-hold.

Asset	Sharpe ratio				Max drawdown (%)	
	B&H	Vol target	p	+ Hawkes gate	B&H	Vol target
<i>Development set</i>						
BTC	0.84	0.78	0.30	0.34	-77.2	-75.9
ETH	0.78	0.71	0.39	0.34	-81.3	-76.0
EUR/USD	-0.11	-0.10	0.30	-0.15	-35.4	-35.3
Gold	0.44	0.47	0.32	0.47	-45.6	-42.7
NIFTY 50	0.72	0.72	0.83	0.72	-38.4	-31.2
Nasdaq-100	0.75	0.84	0.13	0.84	-53.6	-41.6
PAXG	0.95	1.35	0.02	0.90	-29.2	-21.2
S&P 500	0.55	0.58	0.19	0.60	-56.8	-49.3
Treasuries	-0.00	-0.05	0.15	-0.03	-54.2	-53.8
<i>Confirmatory set</i>						
ADA	0.67	0.42	0.00	-0.23	-95.5	-95.0
BNB	1.13	1.08	0.58	0.67	-72.5	-70.9
DAX	0.37	0.38	0.86	0.41	-72.7	-62.7
Emerging mkts	0.23	0.23	0.11	0.13	-43.7	-43.7
FTSE 100	0.27	0.22	0.25	-0.01	-52.6	-46.6
GBP/USD	-0.09	-0.10	0.31	-0.05	-37.5	-37.4
Hang Seng	0.25	0.23	0.63	0.10	-65.2	-58.3
LTC	0.46	0.42	0.52	0.17	-90.4	-85.2
Nikkei 225	0.32	0.32	1.00	0.23	-81.9	-70.1
Oil	0.09	0.08	0.85	0.08	-94.6	-85.2
Russell 2000	0.43	0.39	0.53	0.39	-54.9	-37.5
SOL	0.82	0.83	0.93	0.19	-82.7	-77.2
Silver	0.33	0.34	0.86	0.34	-67.0	-66.7
USD/JPY	0.19	0.22	0.50	0.22	-38.8	-33.7
XRP	0.81	0.60	0.10	0.22	-84.8	-75.1

Table B.9: Unadjusted versus dividend-adjusted prices (volatility: $h = 5$ and $h = 22$; overlay: $h = 5$).

Series	Quantity	Unadjusted	Adjusted	Change
Emerging mkts (h=22)	QLIKE / HAR: Analogue+HAR	1.033	1.033	-0.000
Emerging mkts (h=5)	QLIKE / HAR: Analogue+HAR	1.012	1.007	-0.005
Nasdaq-100 (h=22)	QLIKE / HAR: Analogue+HAR	1.122	1.117	-0.005
Nasdaq-100 (h=5)	QLIKE / HAR: Analogue+HAR	1.018	1.012	-0.006
Russell 2000 (h=22)	QLIKE / HAR: Analogue+HAR	1.172	1.162	-0.010
Russell 2000 (h=5)	QLIKE / HAR: Analogue+HAR	1.010	1.006	-0.004
Treasuries (h=22)	QLIKE / HAR: Analogue+HAR	1.019	1.026	+0.007
Treasuries (h=5)	QLIKE / HAR: Analogue+HAR	1.012	1.010	-0.002
Nasdaq-100	Buy & hold: Sharpe	0.749	0.785	+0.036
Nasdaq-100	Buy & hold: max DD	-0.536	-0.534	+0.001
Nasdaq-100	Vol target: Analogue+HAR: Sharpe	0.839	0.882	+0.043
Nasdaq-100	Vol target: Analogue+HAR: max DD	-0.416	-0.409	+0.007
Treasuries	Buy & hold: Sharpe	-0.001	0.198	+0.198
Treasuries	Buy & hold: max DD	-0.542	-0.484	+0.058
Treasuries	Vol target: Analogue+HAR: Sharpe	-0.045	0.179	+0.224
Treasuries	Vol target: Analogue+HAR: max DD	-0.538	-0.446	+0.092
Russell 2000	Buy & hold: Sharpe	0.426	0.481	+0.055
Russell 2000	Buy & hold: max DD	-0.549	-0.543	+0.006
Russell 2000	Vol target: Analogue+HAR: Sharpe	0.389	0.480	+0.090
Russell 2000	Vol target: Analogue+HAR: max DD	-0.375	-0.352	+0.023
Emerging mkts	Buy & hold: Sharpe	0.235	0.332	+0.098
Emerging mkts	Buy & hold: max DD	-0.437	-0.398	+0.039
Emerging mkts	Vol target: Analogue+HAR: Sharpe	0.228	0.325	+0.098
Emerging mkts	Vol target: Analogue+HAR: max DD	-0.437	-0.393	+0.045

Table B.10: Fixed versus Politis–White bootstrap block lengths.

Set	Test	Cases	Verdict	Fixed	Politis–White	Changed
Development	Volatility MCS (e2)	90	in MCS	61	61	0
Development	VaR/ES MCS (e8)	162	in MCS	135	135	2
Development	Overlay Sharpe test (e5)	63	significant	4	4	0
Confirmatory	Volatility MCS (e2)	150	in MCS	99	105	6
Confirmatory	VaR/ES MCS (e8)	270	in MCS	211	220	11
Confirmatory	Overlay Sharpe test (e5)	105	significant	14	13	3